



Ultrastable and highly sensitive eddy current displacement sensor using self-temperature compensation

Hongbo Wang, Zhihua Feng*

Department of Precision Machinery and Precision Instrumentation, University of Science and Technology of China, Hefei, Anhui 230026, PR China



ARTICLE INFO

Article history:

Received 22 April 2013

Received in revised form

13 September 2013

Accepted 14 September 2013

Available online 23 September 2013

Keywords:

Eddy current sensor (ECS)

Temperature coefficient (TC)

Displacement sensor

AC bridge

Coil impedance

ABSTRACT

This paper proposes a new method to reduce the thermal drift of eddy-current sensors (ECSs) by two orders of magnitude. Theoretical analysis shows that a well-designed bridge will help to decouple two vectors related to the resistance and inductance variations of the sensing coil of ECSs. Experiments show resistance variation has a considerably larger coefficient with temperature change compared to that of inductance variation. Other than being neglected, resistance variation compensates for the influence of temperature on inductance variation, which is used to derive true displacement information. A prototype ECS with high-resolution of sub-nanometer and ultrahigh thermal stability is manufactured and tested. Results show that the thermal drift of the prototype ECS is approximately 2.6 nm/°C, equivalent to 9.7 ppm/°C of the coil's inductance change. This self-temperature compensation method for ECS is simple, low cost, universal, very effective, and has competitive advantages in most applications.

© 2013 Elsevier B.V. All rights reserved.

1. Introduction

Eddy-current sensors (ECSs) are widely used for displacement measurement [1,2]. Generally, an ECS has a sensing coil and a conductive target. It measures the displacement of a target by detecting variations in the sensing coil's impedance (resistance R and inductance L), which varies as a result of the change in distance between the sensing coil and the target [2,3]. Operating on the principle of magnetic induction, ECSs can measure the precise position of a conductive target even when nonmetallic materials such as plastics, nonconductive liquids, and dirt intervene. The sensors are inherently rugged and can operate over a wide temperature range in contaminated environments that would defeat optical, acoustic, and even capacitive sensors. Optical and capacitive devices are usually the best choice for high-resolution displacement sensing [4,5]. However, when the measurement system is configured in contaminated environments, ECS is the only choice. High-resolution ECS has obvious advantages such as large measurement range, wide frequency response, tolerance to dirty environments, and low cost.

However, when ECSs are used as high-resolution displacement sensors with a small measurement range, they can be influenced by temperature fluctuation much easier than capacitive sensors [6]. For example, the SMT9700-15N sensor (KAMAN, USA) has a temperature coefficient (TC) of 0.5 $\mu\text{m}/^\circ\text{C}$ within a 25 μm measuring

range when aluminum is used as target material. This drawback limits the use of the ECS in high-resolution applications, and makes it impossible for the sensor to compete with capacitive sensor within ultrahigh precision measurement applications. This drawback is mainly a result of the displacement information being drowned by the sensing coil's impedance, which includes resistance and inductance. The resistive part varies at a TC rate similar to the electrical resistivity of the coil wire and target material (copper or aluminum). The TC of such material's electrical resistivity can reach as high as 3900 ppm/°C. Thus, ECSs are regarded as displacement sensors that have high resolution but high-temperature drift. Some ECSs with temperature compensation coils have been developed, but these sensors are complex, expensive, as well as hardly promoted and used widely [2,7].

Various researches pointed out that the inductive part has poorer coupling with temperature variation and can be used to obtain more precise displacement information compared with the resistive part [2,8,9]. These studies adapted methods to separate the imaginary part (L) and the real part (R) from the impedance and to solve the problem to a certain extent.

This paper presents a method to reduce the TC of an ECS by two orders of magnitude. The proposed method is based on self-correction and the different sensitivities of inductance and resistance of the sensing coil to displacement and temperature. It is solely based on the processing of impedance signals, does not require additional hardware of the sensor coil or the temperature sensor, and can be applied for any target material. Based on a very simple, but well-designed bridge circuit, R and L can be separated

* Corresponding author. Tel.: +86 55163607894; fax: +86 5513607894.
E-mail address: fff@ustc.edu.cn (Z. Feng).

easily and efficiently. Furthermore, true distance and temperature information can be extracted from the R and L , both of which correspond to the distance and the temperature of the coil-target. In this study, a prototype ECS is manufactured, and the sensitivities of the L and R parts to displacement and to temperature are measured. The characteristics of thermal stability for the prototype ECS that uses self-correction are tested and verified. Finally, the problems encountered in the study are listed, and recommendations to improve the performance of the sensor are proposed.

2. Working principle

As shown in Fig. 1(a), an ECS uses a coil to detect a conductive target. When the sensor coil is driven by an AC current, it generates an alternating magnetic field that induces eddy currents in any nearby conductive object, thus designating the target. The eddy currents generate magnetic field in a direction opposite to that by the sensor coil, which reduces the magnetic flux in the coil, thereby, its inductance. The eddy currents also dissipate energy, which increases the coil's resistance. Moving target changes the coupling between the coil and the target and is reflected in the change of the coil's impedance [10]. The impedance of the sensor coil can be represented as a series LR circuit. The change in both inductance L and resistance R follows the target position. When the target approaches the coil, the inductance usually decreases and the resistance increases. The performance of an ECS can be quantified by the effective impedance of the sensor coil corresponding to the

dynamic range of the target. Effective impedance is a complex function that depends on the characteristics of the sensor coil, target, and coil-target coupling. Thus, the effective impedance of the coil Z_{eff} can be defined as

$$Z_{\text{eff}} = [R_c + R_e(x, \rho, \mu, f)] + j\omega[L_c - L_e(x, \rho, \mu, f)] \quad (1)$$

where R_c and L_c are the inherent resistance and inductance of the coil, respectively, and R_e and L_e are the equivalent resistance and inductance caused by the eddy current in the target. Both R_e and L_e correspond to the coil-target impedance coupling characteristic, which depend on the distance x between the coil and the target, the resistivity ρ and relative permeability μ of the target, and the working frequency f . Coil-target impedance coupling characteristic relies on distance x when the target material and the working frequency are determined. For a non-ferromagnetic material, $\mu = 1$. Resistivity ρ is a function of temperature T . Thus, both R_e and L_e are functions of temperature T and displacement x . The inductance of the coil itself should also be a function of temperature T because the coil's geometric dimensions, which vary following the temperature, correspond to its inductance. Effective impedance can be expressed as

$$Z_{\text{eff}} = [R_c(T) + R_e(x, T)] + j\omega[L_c(T) - L_e(x, T)] \quad (2)$$

where R_c and L_c are the resistance and inductance of the coil respectively, and R_e and L_e are the effective equivalent resistance and inductance from the eddy current with the coupling between coil and target. For a copper coil (copper wire wound coil), the TC of R_c is about 3900 ppm/°C and the TC of L_c is about 10–100 ppm/°C. A low thermal expansion frame such as quartz glass will contribute significantly to the stability of L_c . R_e and L_e depend on the thermal stability of the target's resistivity and frequency, as well as the coupling factor between the coil and the target. For a high-precision ECS probe, the distance between the coil and the target is very small, and aluminum or copper is usually selected as the target material. As a result, the TC of L_e is several hundreds of ppm/°C, but that of R_e is usually more than 2000 ppm/°C. Thus, L_e is more stable than R_e [2].

Given that the ECS signal processing circuit directly detects the total part of the effective impedance without distinguishing the coil part and the part caused by the eddy current [Fig. 1(b)], Eq. (2) can be simplified to

$$Z_{\text{eff}} = R(x, T) + j\omega L(x, T) \quad (3)$$

For a copper coil with a nearby aluminum target, the L is more sensitive to x than the R , but the TC of R is nearly 10 times that of L . The $R(x, T)$ and $L(x, T)$ can be expressed as

$$\begin{cases} R(x, T) = \sum_{i=0}^n \sum_{j=0}^m a_{ij} x^i T^j \\ L(x, T) = \sum_{i=0}^n \sum_{j=0}^m b_{ij} x^i T^j \end{cases} \quad (4)$$

where a_{ij} and b_{ij} are the coefficients of the polynomials.

If the relationships between $R(L)$ and x , as well as $R(L)$ and T are monotonous and unique in the entire measurement range and temperature fluctuation range of the measurement process, then x and T can be rewritten as

$$\begin{cases} x(R, L) = \sum_{i=0}^n \sum_{j=0}^m p_{ij} R^i L^j \\ T(R, L) = \sum_{i=0}^n \sum_{j=0}^m q_{ij} R^i L^j \end{cases} \quad (5)$$

If the coefficients p_{ij} and q_{ij} are known, the temperature drift and the nonlinearity of the ECS can be completely eliminated. For a specific sensor coil and target material, p_{ij} and q_{ij} are constants that can be measured beforehand and used as correction parameters. Large n and m can minimize errors from temperature drift and displacement nonlinearity, but entail lengthy computation time.

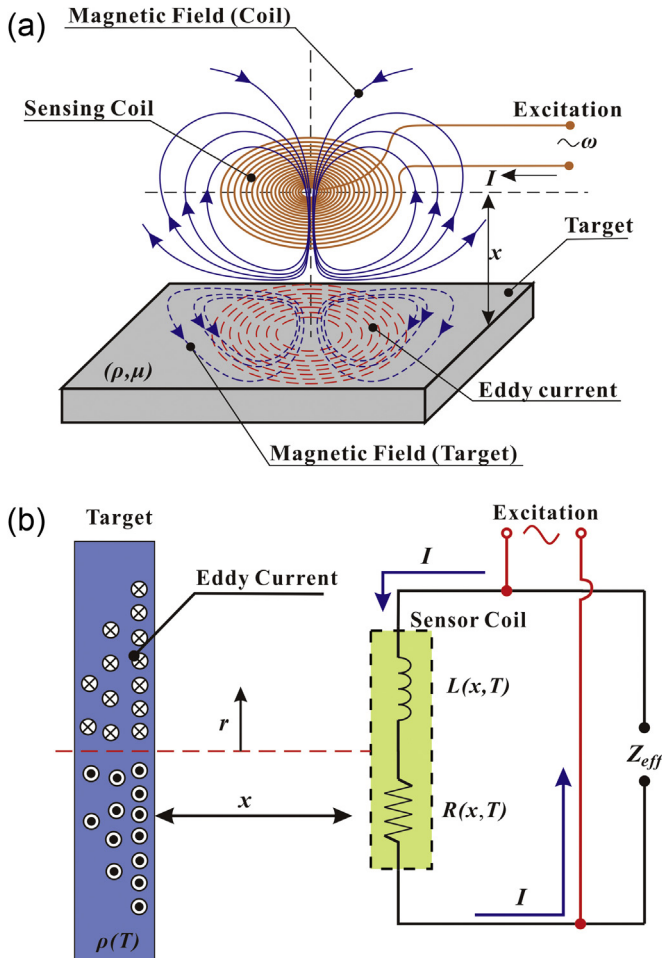


Fig. 1. Eddy-current displacement sensor system: (a) ECS configuration and working principle; (b) equivalent model of ECS.

For a high-resolution ECS system, the typical measurement range of only tens of microns and the temperature fluctuation of less than a few degrees mean the L and R expressions can be simplified to a linear combination of x and T [let $n = m = 1$ in Eq. (4)].

$$\begin{cases} L = K_{11}x + K_{12}T + L_0 \\ R = K_{21}x + K_{22}T + R_0 \end{cases} \quad (6)$$

Solving Eq. (6) yields

$$\begin{cases} x = \frac{K_{22}L - K_{12}R}{K_{11}K_{22} - K_{21}K_{12}} + x_0 \\ T = \frac{K_{11}L - K_{12}R}{K_{21}K_{12} - K_{11}K_{22}} + T_0 \end{cases} \quad (7)$$

Without any additional hardware, the position information x without temperature drift can be simply extracted by adding the R and L signals with different coefficients. Interestingly, even the temperature change can be obtained at the same time.

According to the above analysis, the only problem in correcting thermal drift is how to separate the R and L parts of the coil. Three popular types of circuits are used to process ECS signals: Colpitts oscillator [11], phase or amplitude [12], and AC bridge [13,14]. AC bridge circuits have better linearity, high resolution, and long-term stability. This circuit type is generally composed of an AC bridge, a differential preamplifier, a demodulation circuit, and an output amplifier with zero and gain adjustments [14]. However, all three types measure the impedance of the coil without distinguishing between inductance and resistance.

Several researchers [7,8] used an AC current source to separate R and L , as shown in Fig. 2(a), in which the sensor coil is driven by an AC current source. The R and L parts of the coil impedance are easily separated. Output voltage of the bridge is

$$\tilde{U} = \tilde{I}(\Delta R + j\omega\Delta L) = \tilde{e} \cdot K(\Delta R + j\omega\Delta L) \quad (8)$$

where K is the conversion factor of the voltage–current converter.

When $e(t)$ is used as the reference signal of the lock-in amplifier, the output is proportional to the resistance change ΔR . When another reference signal with a 90° phase shift of $e(t)$ is used, the output is proportional to the inductance change ΔL . Thus, the ΔL and ΔR of the coil can be measured simultaneously using two quadrature lock-in amplifiers. However, as a current source driver cannot be easily implemented for a network that has high-frequency resistance and reluctance, the feasibility of this method is uncertain.

In this study, we use AC voltage sources to separate R and L , and the results are very good. As shown in Fig. 2(b), the output voltage of a bridge activated by an AC voltage source can be calculated by

$$\tilde{U} = \tilde{e}R_s \left[\frac{1}{(R_s + R + \Delta R) + j\omega(L + \Delta L)} - \frac{1}{(R_s + R) + j\omega L} \right] \quad (9)$$

If we select a specific sampling resistance $R_s = \omega L - R$ and assume $\Delta R \ll \omega L$, $\Delta L \ll L$, then Eq. (9) can be simplified to

$$\tilde{U} = \frac{\tilde{e}R_s}{2\omega L} \left(-\frac{\Delta L}{L} + j\frac{\Delta R}{R_s + R} \right) \quad (10)$$

The result is noteworthy because the real part of the output voltage comes from the change of inductance. It indicates that using two lock-in amplifiers with quadrature voltage reference signals can easily separate ΔR and ΔL . The method is also very simple and effective.

3. Device description

The proposed ECS [Fig. 1(a)] has a spiral-monolayered planar coil, a 0.2 mm-thick aluminum plate as the target, and a signal

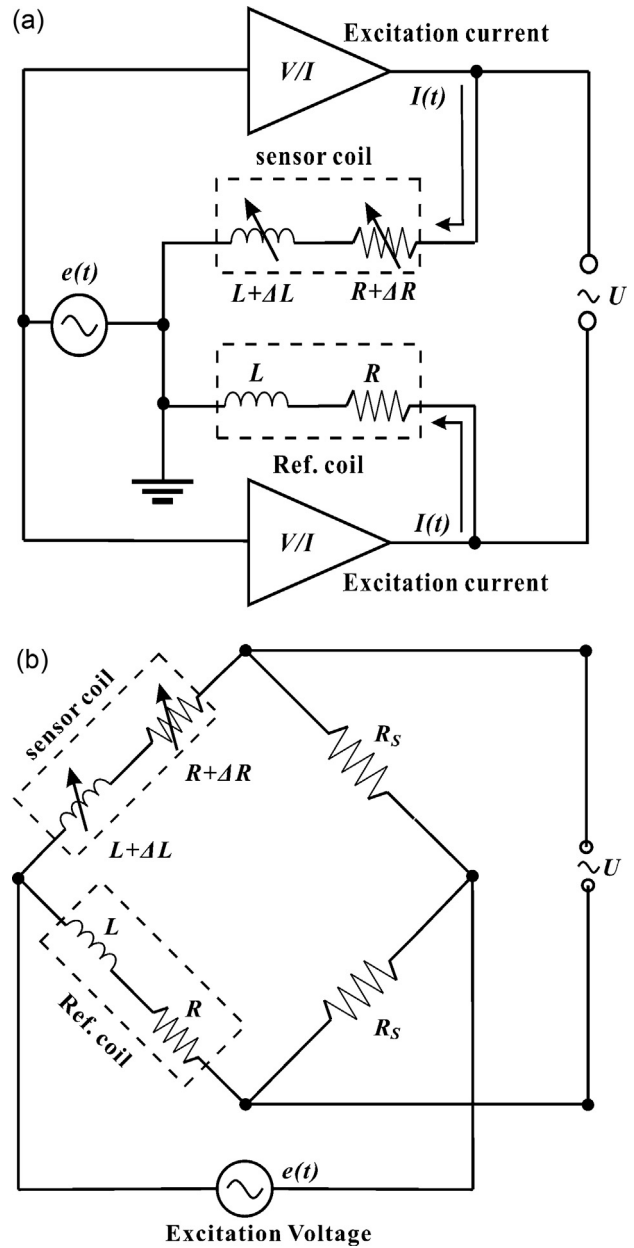


Fig. 2. AC bridge circuit for ECS: (a) current source excitation; (b) voltage source excitation.

processing circuit to demodulate the displacement signal (see Fig. 3). The planar coil is wound with a $50 \mu\text{m}$ OCC wire, consists of 25 turns, and has a 5 mm outer diameter and a 2 mm inner diameter. The coil was excited by a 1 MHz sinusoidal voltage when the ECS was working. The coil plus a cut of Teflon silver-plated coaxial cable has an inductance of $4.56 \mu\text{H}$ and a resistance of 4.16Ω (at 1 MHz). As shown in Fig. 3, the signal processing circuit consists of an excitation voltage source, a well-matched AC bridge, a differential preamplifier (INA103, low noise, low distortion, instrumentation amplifier), two lock-in amplifiers (AD630, high precision balanced modulator/demodulator), and output amplifiers. The reference coil of the AC bridge is wound with a 0.25 mm constantan alloy wire on a quartz glass rod and configured in the signal processing circuit board. The TC of the reference coil's inductance is about $10\text{--}30 \text{ ppm}/^\circ\text{C}$, and the TC of its resistance is about $20 \text{ ppm}/^\circ\text{C}$ to $50 \text{ ppm}/^\circ\text{C}$ because the constantan alloy has a lower TC of resistivity. The two sample resistors R_s have ultrahigh precision and are

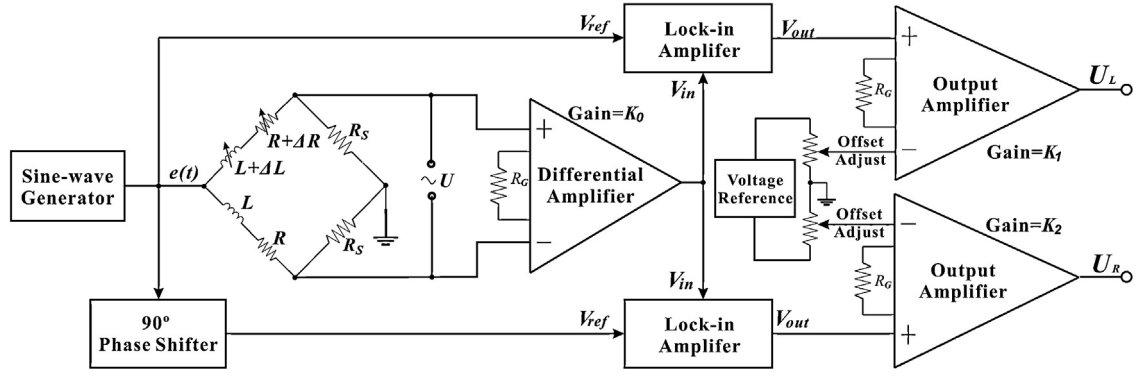


Fig. 3. Schematic of the signal processing circuit of the ECS.

stable with 0.01% accuracy and only 1 ppm/°C temperature drift. In our experimental setup, the distance between the target and the probe coil is around 130 μm ; at this point, the sensing coil has an inductance of 1.7 μH and a resistance of 5.7 Ω . The reference coil is wound carefully to match well with the parameters of the coil target. As a result, the unbalanced output of the AC bridge is very small. As discussed in Section 2, the lock-in amplifier using $e(t)$ as reference signal obtains the inductance change signal U_L , and the quadrature demodulation channel obtains the resistance change signal U_R . The gain and offset of the output amplifier can be adjusted

according to the measurement range. Thus, if the coefficients in Eq. (6) are known, both the displacement signal and the temperature signal can be easily calculated out.

Based on Eq. (6) and the signal processing circuit (Fig. 3), the two channel output voltages are

$$\begin{cases} U_L = \frac{eR_s}{4\omega L} \frac{\Delta L}{L} K_0 \cdot K_1 \\ U_R = \frac{eR_s}{4\omega L} \frac{\Delta R}{R_s + R} K_0 \cdot K_2 \end{cases} \quad (11)$$

where e is the amplitude of the excitation voltage, K_0 is the gain of the differential preamplifier, and K_1 and K_2 are the gains of the two output amplifiers. In our experiments, the value of the parameters are $e = 0.72 \text{ V}$, $L = 1.7 \mu\text{H}$, $R_s = 5.0 \Omega$, $R = 5.7 \Omega$, $f = 1.0 \text{ MHz}$,

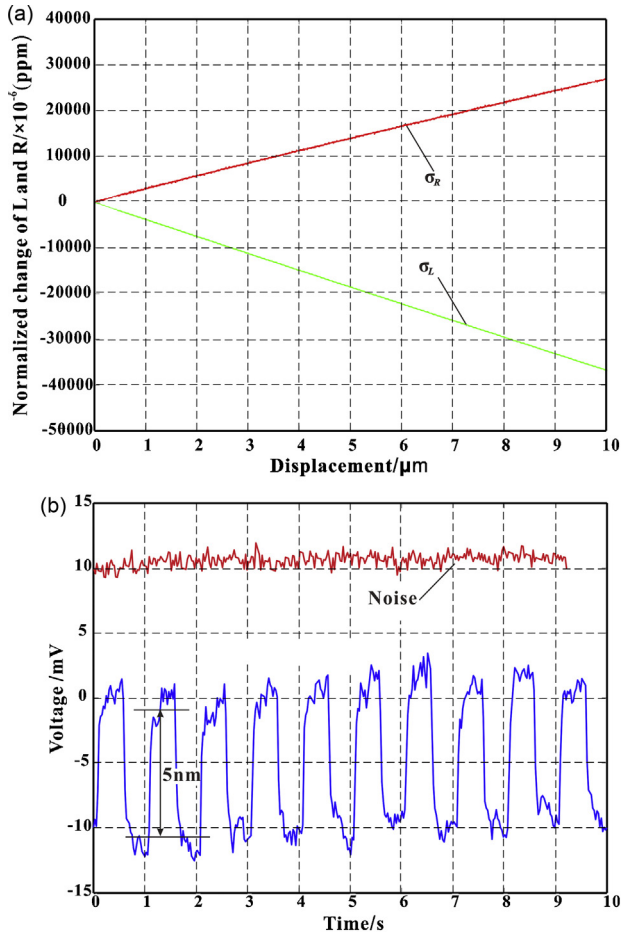


Fig. 4. Response of ECS to displacement: (a) normalized change of L and R to 10 μm displacement change; (b) the output voltage U_L to a 1 Hz-5 nm (peak-to-peak) square-wave distance change and the output noise of U_L .

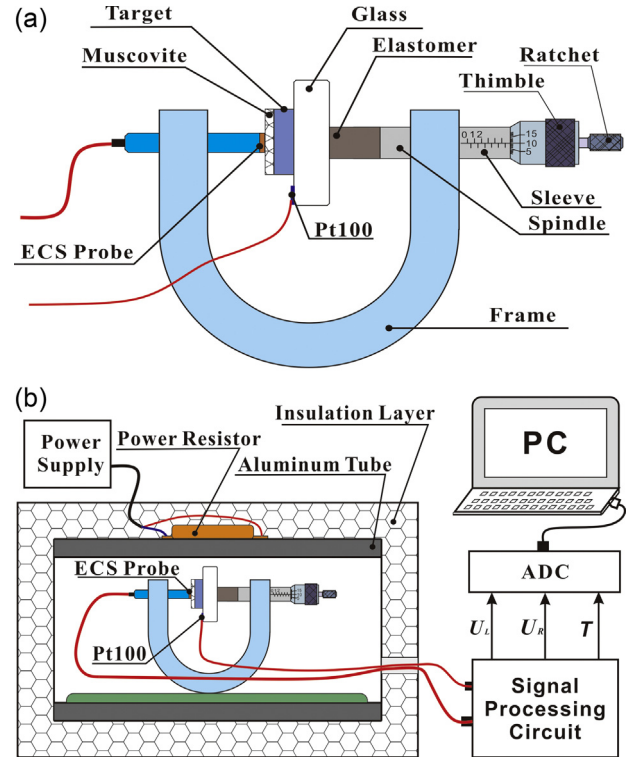


Fig. 5. Temperature drift measurement system: (a) the structure with fixed distance between the coil and target; (b) the configuration of the temperature drift measurement system equipment.

$\omega L = 10.7 \Omega$, $K_0 = 61$, $K_1 = K_2 = 91.83$. Thus, Eq. (11) can be rewritten as

$$\begin{cases} U_L = 476.08 \frac{\Delta L}{L} = 476.08 \sigma_L \\ U_R = 251.26 \frac{\Delta R}{R} = 251.26 \sigma_R \end{cases} \quad (12)$$

where σ_L and σ_R are the normalized change of the inductance and resistance of the coil and are dimensionless quantities. The unit of U_L and U_R is volt (V). Thus, the change of inductance and resistance of the coil can be obtained by calculating the values of U_L and U_R using

$$\begin{cases} \sigma_L = \frac{\Delta L}{L} = \frac{U_L}{476.08} \\ \sigma_R = \frac{\Delta R}{R} = \frac{U_R}{251.26} \end{cases} \quad (13)$$

4. Experiments and results

To obtain the coefficients listed in Eq. (6), the sensitivities of σ_L and σ_R to displacement and temperature were measured.

First, the sensitivities of σ_L and σ_R to displacement were measured. A single-axis piezo-nanopositioner [P-620.1CD-Linearity, Physik Instrumente (PI) GmbH & Co. KG, Germany] was used to generate a linear $10 \mu\text{m}$ displacement. The U_L and U_R of the ECS were measured. The normalized change of the probe's inductance and

resistance are shown in Fig. 4(a). Both U_L and U_R have good linearity, and the nonlinearity errors are less than 0.2%. It should be emphasized that the ECS has good linearity only when the displacement range is very small and the distance between the coil and target are much smaller than the coil's diameter. The sensitivities of σ_L and σ_R to displacement are -3668 and $2680 \text{ ppm}/\mu\text{m}$, respectively. The measuring process was carried out in a thermal insulation box and only lasted 10 s, such that the temperature influence on displacement sensitivity could be ignored. To check the high resolution of the ECS, a 5 nm (peak-to-peak) square-wave (1 Hz) distance change was applied. The response of U_L to the micro-displacement is shown in Fig. 4(b). The figure also shows the noise of U_L . In the figure, the curve shows the capability of the ECS to detect ultra-small displacement down to the sub-nanometer scale. These measurement results were filtered through a 20 Hz low-pass filter to reduce high-frequency noise from the circuit system, environmental vibration interference, and quantization error (0.61 mV for 1 bit) of the ADC.

Measuring the temperature sensitivities of U_L and U_R is a challenging issue, as nearly all metallic and nonmetallic materials have large coefficients of thermal expansion, and the distance between coil probes and targets changes with the temperature change caused by the frame's thermal expansion. Displacement drift from the thermal expansion of the frame may even be larger than the inherent temperature drift of the sensor itself. To measure the thermal drift of the coil-target unit, a setup with very low temperature drift was specially designed [Fig. 5(a)]. A micrometer screw

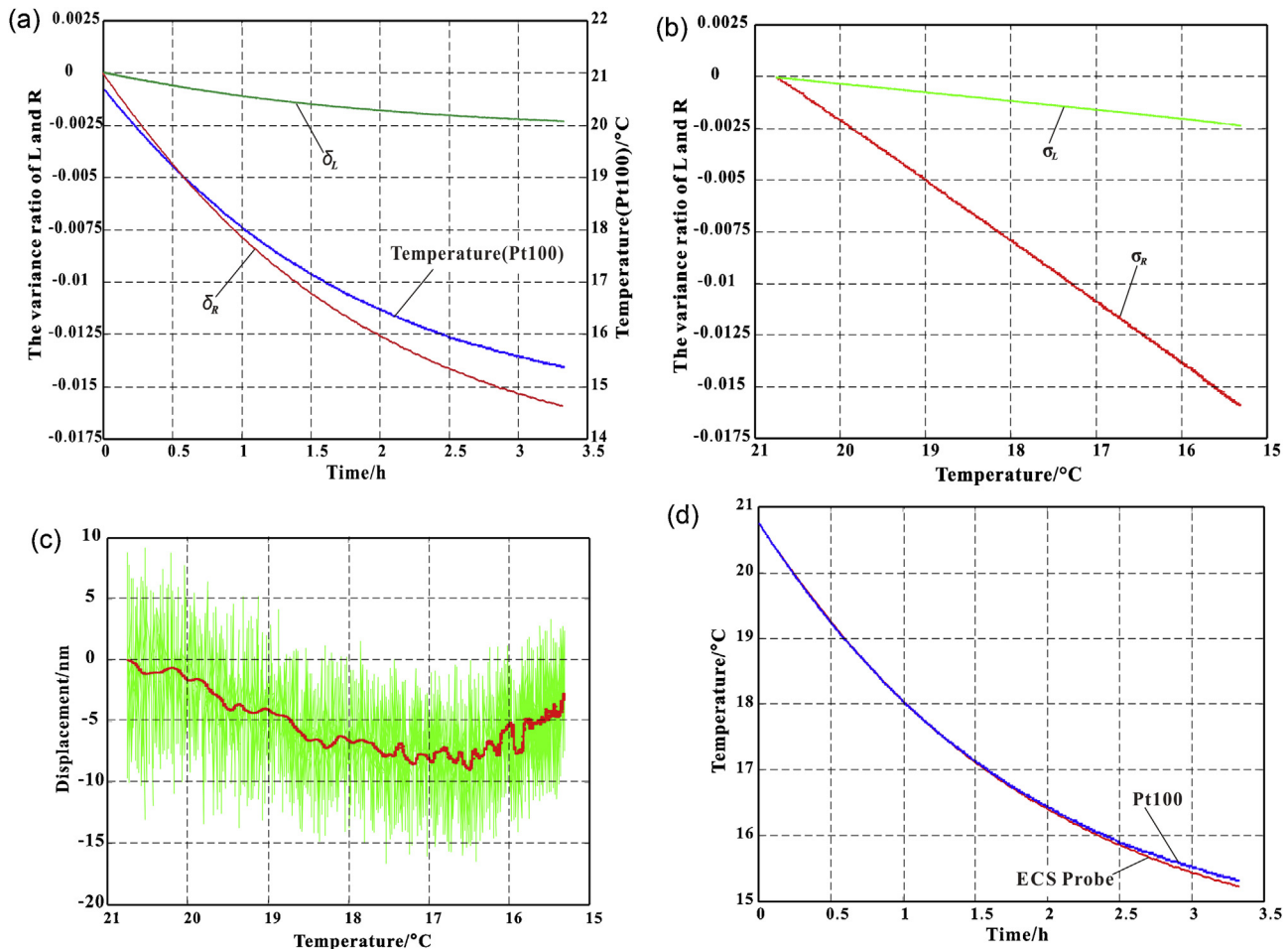


Fig. 6. The temperature drift and self-correction of the ECS probe during a natural cooling process. (a) Normalized change of the coil's inductance and resistance and the temperature within the natural cooling process. (b) Normalized change of the coil's inductance and resistance to temperature. (c) Displacement output of the ECS after self-correction (a filtered smooth curve is also shown). (d) The temperature measured through ECS probe and by Pt100 with time.

gauge was reformed to firmly fix the ECS probe and aligned it coaxially with the spindle. A 0.2 mm-thick aluminum plate was selected as the target plate. A 130 μm -thick muscovite plate was used to determine the distance between the coil and the target surface. When the micrometer screw spindle moved to the left to compress the elastomer at a suitable force, the distance between the coil on the probe and the target surface was fixed to the thickness of the muscovite plate. As the muscovite plate and the probe maintained contact, the thermal expansion of the muscovite plate's thickness was only 1.1 $\text{nm}/^\circ\text{C}$, with a general coefficient of thermal expansion of 9 $\text{ppm}/^\circ\text{C}$ for muscovite. Therefore, the distance between the target surface and the coil on the probe can be regarded as constant during the entire measurement process.

To measure the temperature sensitivities of σ_L and σ_R , the setup was placed in an aluminum tube surrounded by a thermal isolation box [Fig. 5(b)]. A power resistor was used to generate thermal energy and induce a temperature rise of the surrounding environment. A Pt100 was placed near the sensing coil to measure the temperature inside the aluminum tube. This temperature was used to calculate the thermal sensitivity of the two output channels. The test setup was heated and then cooled slowly so the temperature all over the inner space of the aluminum tube could be regarded as the same. The well-designed circuit was placed in an air-conditioned room and regarded as stable.

The system was heated to increase 5°C by the power resistor and then allowed to cool naturally. The curves of relative change rates of L and R of the sensing coil and the temperature with

cooling time are shown in Fig. 6(a). The relationships between σ_L to temperature and σ_R to temperature are shown in Fig. 6(b). The respective coefficients of σ_L and σ_R to temperature are 427.9 and 2938 $\text{ppm}/^\circ\text{C}$. To obtain reliable coefficients, six cycles of heating and cooling were performed. The average coefficients of σ_L and σ_R to temperature are 421.8 and 2881 $\text{ppm}/^\circ\text{C}$ respectively. The mean square deviations of TC of σ_L and σ_R from the six measurements are 9.66 and 62.8 $\text{ppm}/^\circ\text{C}$, respectively. Hence, the correction error of displacement output is about 2.6 $\text{nm}/^\circ\text{C}$.

Substituting the coefficients, Eq. (6) can be rewritten as

$$\begin{cases} \sigma_L = -3.668 \times 10^{-3}x + 0.4218 \times 10^{-3}T + \sigma_{L0} \\ \sigma_R = 2.680 \times 10^{-3}x + 2.881 \times 10^{-3}T + \sigma_{R0} \end{cases} \quad (14)$$

where σ_{L0} and σ_{R0} are constants. Solving Eq. (14) yields the correction equations:

$$\begin{cases} x = 36.06\sigma_R - 240.3\sigma_L + x_0 \\ T = 313.6\sigma_R + 229.1\sigma_L + T_0 \end{cases} \quad (15)$$

where x_0 and T_0 are constants that can be adjusted by the offset nulling of the two output amplifiers according to the target position and temperature point of the ECS probe. Hence, x_0 and T_0 are omitted in the following calculation and discussion. As shown in Fig. 6(c), the fluctuation of displacement output after correction (smooth curve) is approximately 10 nm, with a temperature change of 5°C (less than 2.6 $\text{nm}/^\circ\text{C}$). Note that the high-frequency noise

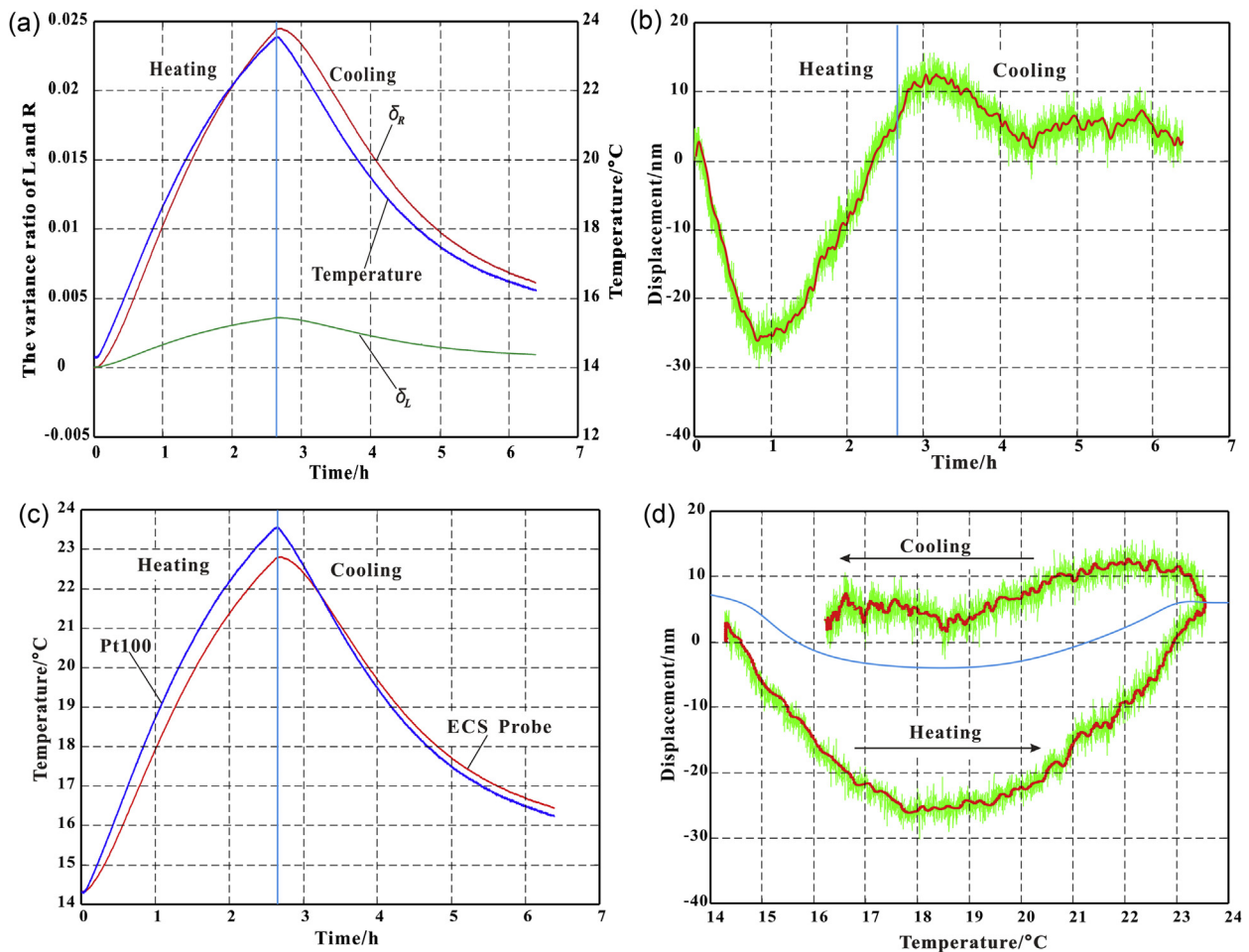


Fig. 7. Temperature drift and self-correction of the ECS probe during heating and cooling. (a) Normalized change of the coil's inductance and resistance and the temperature within the process. (b) Displacement output of the ECS after self-correction with the process (a filtered smooth curve is also shown). (c) Temperature measured through ECS probe and by Pt100 with time. (d) Displacement output of the ECS after self-correction with temperature.

shown in Fig. 6(c) comes mainly from the quantization error of the DAC system (2.44 mV for 1 bit) and the high-frequency noise of the circuit system. The temperatures calculated from σ_L and σ_R with time are shown in Fig. 6(d). The temperature value measured by Pt100 with time is also shown as a reference curve. Clearly, the ECS probe can simultaneously measure the precise displacement and temperature of the probe target. Moreover, the inductance output channel σ_L has a TC as low as $-421.8 \text{ ppm}/^\circ\text{C}$ or an equivalent displacement drift of $115 \text{ nm}/^\circ\text{C}$. However, the output channel σ_R has a TC as high as $2881 \text{ ppm}/^\circ\text{C}$ or an equivalent displacement drift of $1075 \text{ nm}/^\circ\text{C}$. Therefore, even one channel based on inductance change to demodulate displacement information can achieve a relatively low thermal drift, which is only a fifth of the KAMAN's ECS ($500 \text{ nm}/^\circ\text{C}$).

Fig. 7(a) shows σ_L , σ_R and T with time during another heating and cooling process. Fig. 7(b) shows the displacement output after correction. The correction error was only 12 nm during cooling, but reached 30 nm during heating. Fig. 7(c) also shows the temperature data extracted from the two output channels σ_L and σ_R with the measured temperature by Pt100. The curve shows that the ECS probe measured temperature lags behind the Pt100 measured value. The temperature response time of the ECS probe appears to be larger than Pt100 because the probe coil is well packaged by epoxy. According to the Analysis in Section 2, the TC of resistance is mainly determined by the coil temperature, and the TC of inductance is determined by the temperature of the target. Thus, the temperature response time of σ_R is larger than that of σ_L . This finding means that the temperature response of σ_L and σ_R will be unsynchronized when the temperature changes too fast. As a result, the correction error will increase, as shown by the heating process in Fig. 7(b). Therefore, the rate of temperature change should be less than $1^\circ\text{C}/\text{h}$ to ensure the accuracy of correction. Fig. 7(d) shows the displacement output after correction with temperature.

5. Discussion and conclusion

The results of this study prove the validity of the proposed method. The ECS probe can detect the precise displacement and temperature without any additional hardware of the probe by using two channel quadrature lock-in amplifiers to simultaneously detect inductance and resistance change of the coil. Results show the self-correcting ECS can achieve an ultralow thermal drift of $2.6 \text{ nm}/^\circ\text{C}$, which is equivalent to $9.66 \text{ ppm}/^\circ\text{C}$ of the coil's inductance and is comparable to the capacitive sensor. In measuring quasi-static precision displacement, the proposed ECS has advantages of high environmental tolerance and low cost over the capacitive sensor. The limitation of this correction method is that the rate of temperature change should be less than $1^\circ\text{C}/\text{h}$, which is a necessary for most (sub-) nanometer resolution displacement measurements. Solving this limitation and expanding the applications of the probe can be achieved by improving the process of the probe package to obtain similar thermal response times for σ_L and σ_R . For different target materials, the correction coefficients must be remeasured and the parameters of the AC bridge selected to meet the conditions discussed in Section 2. Therefore, variable precision and high thermal stability inductors and resistor series connected as a reference coil is required. This LR series has better performance than the reference coil wound with constantan alloy wire in our ECS signal processing circuit. It can also match the AC bridge for different target materials or measuring points by adjusting the value of the reference inductors and resistors. With these improvements, the self-correcting ECS probe proposed in this paper can adapt to any target material and measuring point without any additional hardware of the probe or any hardware change of the signal processing circuit. The probe

only needs to adjust the reference inductors and resistors, remeasure the correction coefficient, and adjust the scale factor of the output amplifier. When wide temperature range and large measuring range are needed, polynomials Eq. (5) can be used to correct the nonlinearity and the temperature drift.

This study proposes a universal method to reduce the thermal drift of high-precision ECS by a factor more than 100. The proposed self-correcting ECS has sub-nano resolution and ultrahigh thermal stability. The high-resolution ECS with self-correction of thermal drift can be used in any high-precision noncontact position, displacement, and vibration measurement applications in contaminated environments. It has competitive advantages over the capacitive sensor in most applications. Furthermore, this method can be applied in thermal drift correction of other ECS probe for electrical resistivity measurement, or thickness measurement. In fact, the self-temperature compensation technique suggested in this paper can be applied to any measurement systems where the transducer has more than one temperature-dependent transfer characteristic for the same measurand. Each transfer characteristic provides an equation, which is corresponding to temperature and the measurand. Based on the two independent equations, one can exclude temperature from the system of two equations and obtain a temperature-independent measurand modal. Therefore, this method could be applied to many other sensor systems to obtain the temperature-independent result of the measurand.

References

- [1] Eddy-Current Displacement Sensors Overview, Application Notes, Lion Precision, 2011.
- [2] S.D. Roach, Designing and building an eddy-current position sensor, *Sensors* 15 (9) (1998) 1–16.
- [3] T. Hitz, S.D. Welsby, True Position Measurement with Eddy Current Technology, 1997.
- [4] Differences Between Capacitive and Eddy-Current Sensors, TechNote, Lion Precision, 2012.
- [5] M.R. Nabavi, S.N. Nihtianov, Design strategies for eddy-current displacement sensor systems: review and recommendations, *IEEE Sensors Journal* 12 (December (12)) (2012) 3346–3355.
- [6] A.J. Fleming, A review of nanometer resolution position sensors: operation and performance, *Sensors and Actuators A: Physical* 190 (2013) 106–126.
- [7] Q. Li, F. Ding, Novel displacement eddy current sensor with temperature compensation for electrohydraulic valves, *Sensors and Actuators A: Physical* 122 (1) (2005) 83–87.
- [8] Danielson T. Multi-parameter eddy current measuring system with parameter compensation technical field: U.S. Patent 5,541,510[P]. 1996-7-30.
- [9] Kustra T.A., Caffarel A.J. Method and apparatus for correcting eddy current signal voltage for temperature effects: U.S. Patent 4,893,079[P]. 1990-1-9.
- [10] Y. Le Bihan, Study on the transformer equivalent circuit of eddy current non-destructive evaluation, *NDT & E International* 36 (5) (2003) 297–302.
- [11] X.L. Chen, T.H. Ding, Flexible eddy current sensor array for proximity sensing, *Sensors and Actuators A: Physical* 135 (March (1)) (2007) 126–130.
- [12] G.Y. Tian, Z.X. Zhao, R.W. Baines, The research of inhomogeneity in eddy current sensors, *Sensors and Actuators A: Physical* 69 (August (2)) (1998) 148–151.
- [13] W. Lord, Y.S. Sun, S.S. Udpa, et al., A finite element study of the remote field eddy current phenomenon, *IEEE Transactions on Magnetics* 24 (1) (1988) 435–438.
- [14] Inductive Technology Handbook, Application Handbook, Kaman Aerospace Corporation, 2011.

Biographies

Hongbo Wang was born in 1988. He received the B.E. degree from the University of Science and Technology of China (USTC), Hefei, China, in 2010, where he is currently a Ph.D. student in the Department of Precision Machinery and Precision Instrumentation. His research interests include magnetometers and magnetic sensor, piezoelectric actuators and sensors, eddy current and capacitive displacement sensors, and measurement and control techniques and instruments.

Zhihua Feng was born in 1964. He received the B.E., M.E., and Ph.D. degrees from the University of Science and Technology of China (USTC), Hefei, China, in 1987, 1990, and 2005, respectively. He is currently a professor in the Department of Precision Machinery and Precision Instrumentation, USTC. His research interests include smart actuators and sensors, energy harvesting, power electronics, and robotics.